Healthcare Operations and Performance

Business Insights Report for HealthConnect Rwanda leadership

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Data	27 HealthConnect datasets: 20,000 patients, 5,000 detailed appointments, 60,000 bills, 20 hospitals (HF001 to HF020), 1,250 staff. Data covers January to December 2026.
Money	All amounts are in Rwandan francs (RWF).

1. The short story

HealthConnect runs 20 hospitals that look very alike on paper: each has about 17 to 18 doctors, 35 nurses, and sees roughly 220 to 280 appointments. Patients pay most of what they are billed (90.1%). Yet three things stand out and each one points to money or care that can be recovered:

- **The same staff team serves very different workloads.** Hospitals HF013, HF014 and HF011 handle about 16 appointments per doctor. HF004, HF005 and HF007 handle about 12. Staffing was not matched to demand.
- **The busiest hospitals are also among the least liked.** HF013 and HF014 are the two busiest hospitals and are among the lowest rated (2.77 out of 5). Overall, 4 in 10 patient ratings are poor (1 or 2 stars).
- **RWF 920 million is still unpaid, and half of insurance claims are still pending.** That is RWF 2.44 billion of claims waiting for approval.

Key numbers	Value	Key numbers	Value
Registered patients	20,000	Total billed (60,000 bills)	RWF 9.31 billion
Appointments (detailed sample)	5,000	Collected	RWF 8.39 billion (90.1%)
Hospitals / staff / beds	20 / 1,250 / 2,000	Still unpaid	RWF 919.9 million (9.9%)
Admission rate	11.7% (586 of 5,000)	Insurance claims approved	50.0% (21,008 of 42,036)
Average stay in hospital	5.5 days	Average patient rating	2.99 out of 5
Patients aged over 65	5,487 (27%)	Patients with a chronic condition	14,929 (74.6%)

2. What we did and how to read the numbers

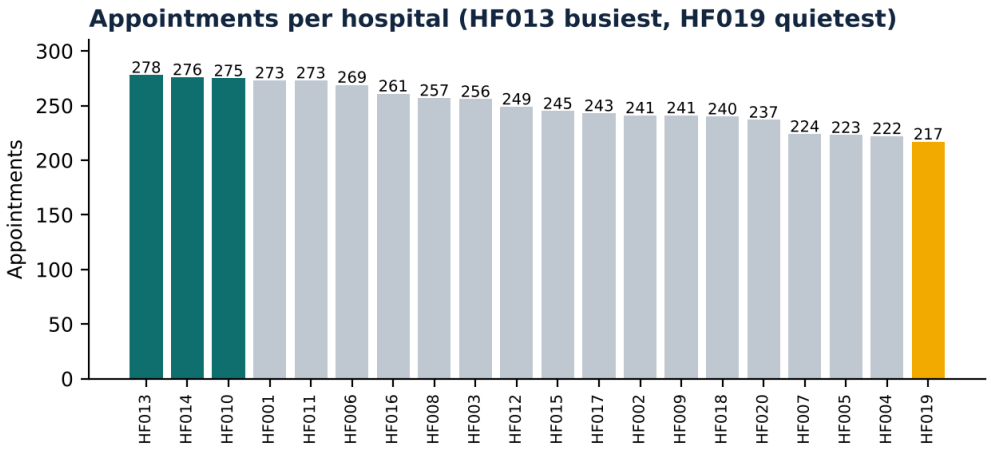
The data was loaded into a PostgreSQL database, cleaned and checked (48 checks, recorded in the data quality log), then analysed with 40 SQL questions and shown in a six-page Power BI dashboard. One important thing to know: **only 5,000 appointments have full details (hospital, department, patient). Bills, diagnoses and feedback records refer to up to 60,000 appointments, but 92% of those appointments are missing from the appointment list.** So:

- **Totals** (all patients, all bills, all diagnoses, all ratings) use every record.
- **Results by hospital, department and province** use only the 5,000 appointments that can be traced. They show real patterns, but about 8% of the bills and 8% of the ratings.
- The data has **no hospital names, types or locations**, only the codes HF001 to HF020. Provinces therefore refer to **where the patient lives**, not where the hospital stands.

3. Answers to the ten business questions

Q1. Which hospitals see the most patients?

HF013 (278 appointments), HF014 (276) and HF010 (275) are the busiest; HF001 and HF011 follow with 273 each. HF019 is the quietest (217). The gap between busiest and quietest is only about 28%, so no hospital is clearly overloaded or empty.



Q2. Which departments are busiest?

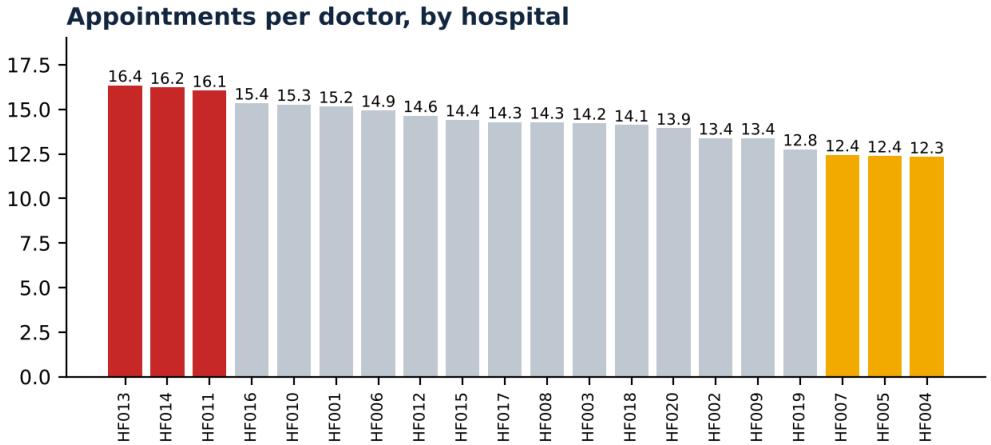
Cardiology (544 appointments), Radiology (525) and Obstetrics and Gynaecology (511) lead. ICU is the smallest (467). Cardiology is also the top earner in the traceable bills (RWF 86.2 million).

Q3. Which diseases are becoming more common?

The five recorded diagnoses are almost equally common across all 60,000 records: Gastritis 12,208, Diabetes 12,083, Hypertension 12,075, Malaria 11,878 and Upper Respiratory Infection 11,756. In the traceable appointments, **hypertension rose from 253 cases in the third quarter to 277 in the fourth, and upper respiratory infections reached their highest level in the fourth quarter (273)**. Malaria went the other way (251 in the second quarter, 211 in the fourth). The fourth quarter includes appointments dated after the analysis. The changes are small, so treat them as a watch-list, not a firm trend. One third of all diagnoses (33.3%) are marked severe.

Q4. Is staff shared out efficiently?

No. Every hospital has the same 35 nurses and 17 or 18 doctors, whatever its patient load. That produces a wide spread in workload (chart below).



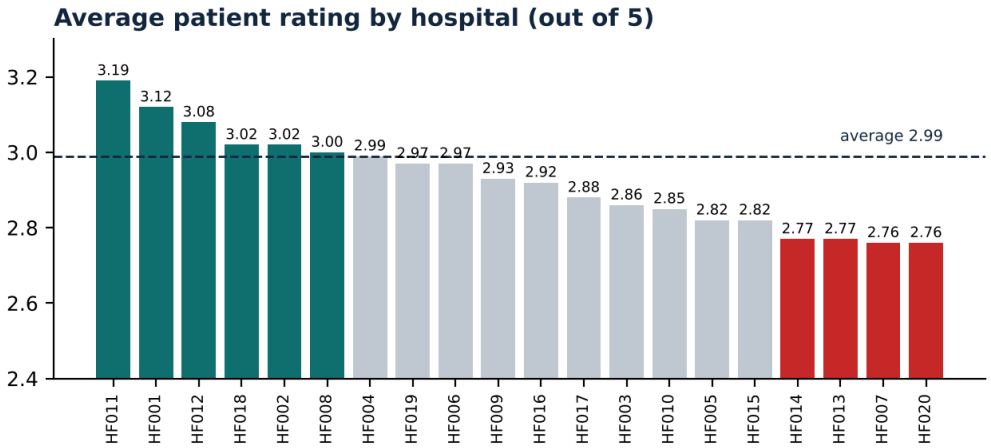
Q5. Which insurers bring in the most?

Insurer	Claims	Amount claimed (RWF)	Approved	Value approved (RWF)
Radiant	10,644	1.237 billion	50.1%	619.8 million
Private	10,508	1.217 billion	49.3%	595.2 million
RSSB	10,511	1.215 billion	50.9%	623.8 million
MMI	10,373	1.213 billion	49.6%	604.4 million

Radiant claims the most, RSSB has the highest approval rate and approved value (RWF 623.8 million). The four insurers are within 2% of each other, so none is a clear problem or star. The bigger point is that **no claim has been rejected; 50% are still marked Pending.**

Q6. Where are patients happiest?

HF011 (3.19), HF001 (3.12) and HF012 (3.08) have the best ratings. HF020 and HF007 (2.76) and HF013 and HF014 (2.77) have the lowest. By department, Emergency (3.02) is best and ICU (2.76) is worst. Ratings are spread evenly: 20% of patients give each score from 1 to 5, which means the average hides a big split between happy and unhappy patients.

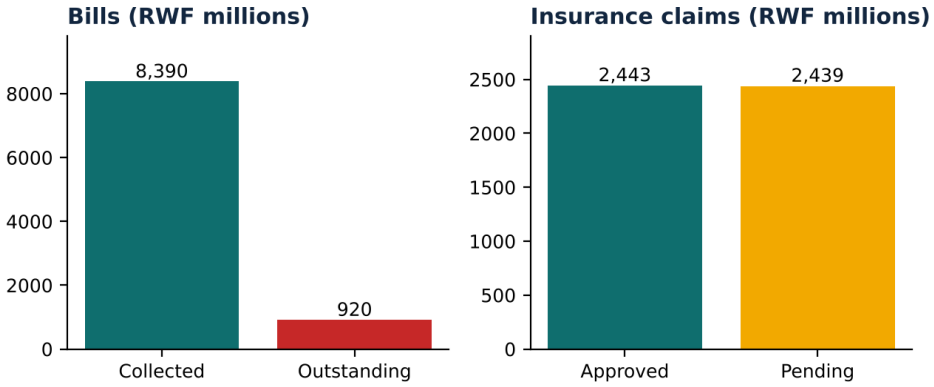


Q7. Which hospitals may need expansion?

HF013 and HF014 are the strongest case: top-two volume, 16 or more appointments per doctor, and some of the lowest ratings. **HF010** (third busiest, rating 2.85) and **HF011** (16 per doctor, but the best ratings, so it is coping) come next. Beds are not the problem: the 2,000 beds are used for about 27,400 bed-days out of 730,000 possible (estimated usage 3.8%). Expansion should mean **more doctors and consulting capacity, not more beds**.

Q8. Where can costs be reduced or money recovered?

Three areas: (1) **RWF 919.9 million unpaid** on 11,887 bills; unpaid amounts are almost the same across cash, card, mobile money and insurance, so the problem is follow-up, not one payment method. (2) **RWF 2.44 billion of claims is pending**. (3) **Billing records are unreliable:** 36,052 of the 60,000 bills carry a status label (Paid, Pending, Partially Paid) that contradicts the payment actually received. Fixing this will help finance see what is really owed.

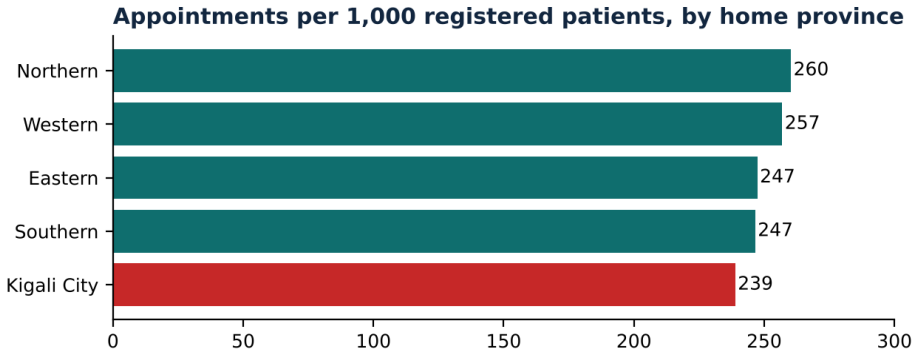


Q9. Which services earn the most?

The data has no service on a bill, so the type of visit is used. Follow-up visits (33.8%), consultations (33.4%) and emergencies (32.8%) earn almost the same. By department, Cardiology (RWF 86.2 million), Radiology (82.0 million) and Laboratory (78.6 million) earn the most from traceable bills. The average bill is RWF 155,157. Monthly revenue is uneven: the top months are September (RWF 145.8 million) and May (134.1 million); the lowest is December (66.7 million).

Q10. Which regions need investment?

Patients are spread evenly across the five provinces (3,923 to 4,091 each). Use of hospitals is not: patients from **Kigali City use hospitals least (239 appointments per 1,000 patients)** and Southern (247) and Eastern (247) follow, while Northern (260) and Western (257) use them most. Western patients also produce the most revenue (RWF 162.0 million of traceable bills) and Kigali City the least (146.1 million). Kigali City, Southern and Eastern deserve a closer look at access and outreach.



4. Recommendations for leadership

What to do	Why	Where to start
1. Re-balance doctors across hospitals	Workload ranges from 12 to 16 appointments per doctor with the same staffing	Move or add doctors at HF013, HF014, HF011; review HF004, HF005, HF007
2. Launch a payment follow-up drive	RWF 920 million unpaid (9.9% of billing)	Start with the largest unpaid bills; send reminders for the 11,887 bills
3. Clear the insurance claim backlog	21,028 claims (RWF 2.44 billion) pending; no rejections recorded	Agree a 30-day approval target with each insurer; start with RSSB and Radiant
4. Run a patient-experience programme	40.5% of ratings are 1 or 2 stars; ICU and the busiest hospitals score lowest	HF013, HF014, HF007, HF020 and ICU; find out why with short follow-up surveys
5. Fix the data	Facility names, links between records, bill status, phone numbers and cancellations are missing or wrong	See the list in section 5; assign an owner and a deadline
6. Look at access in Kigali City, Southern and Eastern	Lowest hospital use per patient	Community outreach and a survey of why patients do not attend

5. What the data could not tell us (limitations)

- **Hospital names, types and locations are missing** (only HF001 to HF020). Rankings by hospital type or ownership, and maps of hospitals, were not possible.
- **Only 5,000 appointments are detailed.** Hospital, department and province results rest on these. We recommend loading the full list of 60,000 appointments.
- **Appointments run to 28 December 2026**, beyond the date of this analysis; 916 are in the future. They are shown but labelled.
- **There is no cancellation or no-show field,** so cancellation rates cannot be measured. All beds are marked "Available", so bed use is an estimate from hospital stays.
- **The Revenue Summary (RWF 1.13 billion) does not match the bills (RWF 9.31 billion).** Bills are used for totals; the Revenue Summary only for month-by-month movement.
- **Phone numbers are damaged** (saved in scientific notation) and were not used.

6. Next steps

Share this report with the Chief Medical Officer and the Finance Director; confirm the data fixes in section 5; and refresh the dashboard once the full appointment list is available. All work is repeatable: the SQL scripts, the database backup, the Power BI data files and the DAX measures are in the project folder.